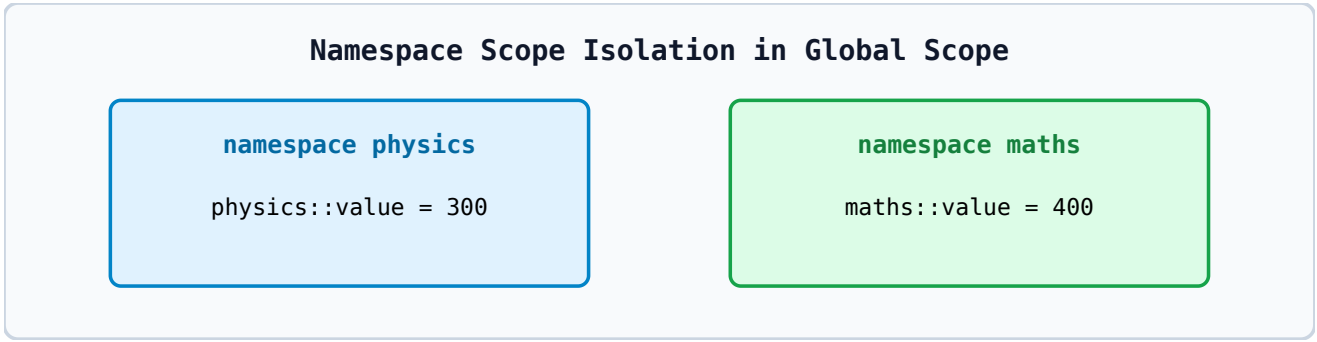
19 NAMESPACES AND PREPROCESSOR • THEORY

01 namespaces and scope resolution

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

01 — Namespaces, `using` Declarations, and `using` Directives



Namespace Scopes

- **Namespace Definition**: Groups identifiers (`classes` , `functions` , `constants`) into a named scope (`namespace math_dept { int code = 101; }`) to prevent global naming collisions.
- **Explicit Qualification**: `math_dept::code`
- `using` **Declaration**: `using math_dept::code;` imports only that single name.
- `using` **Directive**: `using namespace math_dept;` makes all names in `math_dept` visible.